Unit - II

Partial Differentiation:

Consider $z = f(x, y)$, the first order partial derivative of z w.r.t 'x' is $\frac{\partial z}{\partial x}$ or z_x or $\frac{\partial f}{\partial x}$, it is the derivative of z w.r.t 'x' assuming 'y' as constant. The first order partial derivative of z w.r.t 'y' is $\frac{\partial z}{\partial y}$ or z_y or $\frac{\partial f}{\partial y}$, it is the derivative of z w.r.t 'y' assuming 'x' as constant.

$$\frac{\partial z}{\partial x} = z_x = f_x = \lim_{\delta x \to 0} \left[\frac{f(x+\delta x, y) - f(x, y)}{\delta x} \right]$$

$$\frac{\partial z}{\partial y} = z_y = f_y = \lim_{\delta y \to 0} \left[\frac{f(x, y+\delta y) - f(x, y)}{\delta y} \right]$$

Scanned with CamScanner

Second order partial derivatives of $z = f(x, y)$ are

$$\frac{\partial^2 z}{\partial x^2} = z_{xx} = f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)$$

$$\frac{\partial^2 z}{\partial y^2} = z_{yy} = f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)$$

$$\frac{\partial^2 z}{\partial x \partial y} = z_{xy} = f_{xy} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$$

$$\frac{\partial^2 z}{\partial y \partial x} = z_{yx} = f_{yx} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$$

Note that $z_{xy} = z_{yx}$

Consider $z = g(x, y, u, v)$, then first order partial derivatives of z are $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.

Eg 1: Consider $z = x^3 + y^3 - 3axy$

$$z_x = \frac{\partial z}{\partial x} = 3x^2 - 3ay$$

$$z_y = \frac{\partial z}{\partial y} = 3y^2 - 3ax$$

$$z_{xx} = 6x, \quad z_{yy} = 6y, \quad z_{xy} = -3a, \quad z_{yx} = -3a$$

Scanned with CamScanner

Eg 2: Let $z = e^{ax+by}$

$$\frac{\partial z}{\partial x} = a\left(e^{ax+by}\right)$$

$$\frac{\partial z}{\partial y} = b\left(e^{ax+by}\right)$$

$$z_{xx} = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial x}\left(a e^{ax+by}\right) = a^2 e^{ax+by}$$

$$z_{yy} = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial y}\left(b e^{ax+by}\right) = b^2 e^{ax+by}$$

$$z_{xy} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}\left(b e^{ax+by}\right) = ab e^{ax+by}$$

$$z_{yx} = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}\left(a e^{ax+by}\right) = ab e^{ax+by}$$

$$z_{xy} = z_{yx}$$

Eg 3: Let $u = \sin(2x+5y)$

$$u_x = \frac{\partial u}{\partial x} = \left[\cos(2x+5y)\right] 2$$

$$u_y = \frac{\partial u}{\partial y} = \left[\cos(2x+5y)\right] 5$$

Scanned with CamScanner

$$u_{xx} = \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left[2\cos(2x+5y)\right]$$

$$= 2\left[-\sin(2x+5y)\right]2$$

$$= -4\sin(2x+5y)$$

$$u_{yy} = \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial y}\left[5\cos(2x+5y)\right]$$

$$= 5\left[-\sin(2x+5y)\right]5$$

$$= -25\sin(2x+5y)$$

$$u_{xy} = \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial x}\left[5\cos(2x+5y)\right]$$

$$u_{xy} = 5\left[-\sin(2x+5y)\right]2 = -10\sin(2x+5y$$

$$u_{yx} = \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial y}\left[2\cos(2x+5y)\right]$$

$$u_{yx} = 2\left[-\sin(2x+5y)\right]5$$

$$u_{yx} = -10\left[\sin(2x+5y)\right]$$

$$u_{xy} = u_{yx}$$

Scanned with CamScanner

Eg 4: $u = e^x \cos y$

$$u_x = (\cos y)\, e^x$$

$$u_y = (e^x)(-\sin y)$$

$$u_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left[e^x \cos y\right] = (\cos y)\, e^x$$

$$u_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial y}\left[-\sin y\ e^x\right] = (-\cos y)\, e^x$$

$$u_{xy} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial x}\left[-\sin y\ e^x\right] = (-\sin y)\, e^x$$

$$u_{yx} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial y}\left[\cos y\ e^x\right] = (-\sin y)\, e^x$$

$$\Rightarrow u_{xy} = u_{yx}$$

Eg 5: $z = \tan^{-1}(y/x)$

$$\frac{\partial z}{\partial x} = \left[\frac{1}{1+\left(\frac{y}{x}\right)^2}\right] y \left(\frac{-1}{x^2}\right) = \left[\frac{1}{\frac{x^2+y^2}{x^2}}\right]\left[\frac{-y}{x^2}\right]$$

$$\frac{\partial z}{\partial x} = \frac{-y}{x^2+y^2}$$

$$\frac{\partial z}{\partial y} = \left[\frac{1}{1+\left(\frac{y}{x}\right)^2}\right]\left(\frac{1}{x}\right)(1) = \left[\frac{1}{\frac{x^2+y^2}{x^2}}\right]\left(\frac{1}{x}\right) = \frac{x}{x^2+y^2}$$

Scanned with CamScanner

$$z_{xx} = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{-y}{x^2+y^2}\right)$$

$$= (-y)\frac{\partial}{\partial x}\left(\frac{1}{x^2+y^2}\right)$$

$$= (-y)\left[\frac{-1}{(x^2+y^2)^2}\right](2x+0)$$

$$= \frac{2xy}{(x^2+y^2)^2}$$

$$z_{yy} = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)$$

$$= \frac{\partial}{\partial y}\left(\frac{x}{x^2+y^2}\right) = x\frac{\partial}{\partial y}\left(\frac{1}{x^2+y^2}\right)$$

$$= x\left[\frac{-1}{(x^2+y^2)^2}\right](0+2y)$$

$$= \frac{-2xy}{(x^2+y^2)^2}$$

Scanned with CamScanner

$$z_{xy} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right)$$

$$= \left[\frac{(x^2+y^2)(1) - x(2x+0)}{(x^2+y^2)^2}\right]$$

$$z_{xy} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$z_{yx} = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}\left(\frac{-y}{x^2+y^2}\right)$$

$$z_{yx} = \frac{(x^2+y^2)(-1) - (-y)(0+2y)}{(x^2+y^2)^2}$$

$$z_{yx} = \frac{-x^2-y^2+2y^2}{(x^2+y^2)^2}$$

$$z_{yx} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\therefore \quad z_{xy} = z_{yx}$$

Scanned with CamScanner

6Q) If $u = \log(x^2+y^2)$, then show that $u_{xy} = u_{yx}$.

→ $u = \log(x^2+y^2)$

$$u_x = \frac{\partial u}{\partial x} = \left(\frac{1}{x^2+y^2}\right)(2x)$$

$$u_y = \frac{\partial u}{\partial y} = \left(\frac{1}{x^2+y^2}\right)(2y)$$

$$u_{xy} = \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial x}\left[\frac{2y}{x^2+y^2}\right]$$

$$u_{xy} = \frac{(x^2+y^2)(0) - (2y)(2x+0)}{(x^2+y^2)^2}$$

$$u_{xy} = \frac{-4xy}{(x^2+y^2)^2}$$

$$u_{yx} = \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial y}\left(\frac{2x}{x^2+y^2}\right)$$

$$u_{yx} = \frac{(x^2+y^2)(0) - (2x)(0+2y)}{(x^2+y^2)^2}$$

$$u_{yx} = \frac{-4xy}{(x^2+y^2)^2} \quad \therefore u_{xy} = u_{yx}$$

Scanned with CamScanner

classmate
Date
Page

7) If $u = \frac{e^{x+y}}{e^x + e^y}$, then show that $u_x + u_y = u$.

→ $u = \frac{e^{x+y}}{e^x + e^y} = \frac{e^x e^y}{e^x + e^y}$

$u_x = \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left[\frac{e^x e^y}{e^x + e^y}\right] = e^y \frac{\partial}{\partial x}\left[\frac{e^x}{e^x + e^y}\right]$

$u_x = e^y \left[\frac{(e^x + e^y)(e^x) - e^x(e^x + 0)}{(e^x + e^y)^2}\right]$

$u_x = e^y \left[\frac{e^x e^x + e^y e^x - e^x e^x}{(e^x + e^y)^2}\right] = \frac{e^y e^y e^x}{(e^x + e^y)^2}$

$u_y = \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left(\frac{e^x e^y}{e^x + e^y}\right) = e^x \frac{\partial}{\partial y}\left(\frac{e^y}{e^x + e^y}\right)$

$u_y = e^x \left[\frac{(e^x + e^y)(e^y) - e^x(0 + e^y)}{(e^x + e^y)^2}\right]$

$u_y = e^x \left[\frac{e^x e^y + e^y e^y - e^x e^y}{(e^x + e^y)^2}\right]$

$u_y = \frac{e^x e^x e^y}{(e^x + e^y)^2}$

$u_x + u_y = \frac{e^y e^y e^x}{(e^x + e^y)^2} + \frac{e^x e^x e^y}{(e^x + e^y)^2} = \frac{e^x e^y (e^y + e^x)}{(e^x + e^y)^2} = u$

Scanned with CamScanner

8 Q) If $z = f(x+ct) + \phi(x-ct)$, then show that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$.

→ $z = f(x+ct) + \phi(x-ct)$

$z_x = \frac{\partial z}{\partial x} = f'(x+ct) + \phi'(x-ct)$

$z_{xx} = \frac{\partial^2 z}{\partial x^2} = f''(x+ct) + \phi''(x-ct)$

$z_t = \frac{\partial z}{\partial t} = [f'(x+ct)]c + [\phi'(x-ct)](-c)$

$z_{tt} = \frac{\partial^2 z}{\partial t^2} = [f''(x+ct)]c^2 + [\phi''(x-ct)](-c)^2$

$z_{tt} = c^2 [f''(x+ct) + \phi''(x-ct)]$

$\Rightarrow z_{tt} = c^2 z_{xx}$

—

Scanned with CamScanner

Homogenous Function :

A Homogenous function of nth degree is of the form $z = x^n f(y/x)$

or $z = y^n g(x/y)$

Example:

(1) $z = 5x^3 + 2x^2y + xy^2$

$$z = x^3\left[5 + \frac{2x^2y}{x^3} + \frac{xy^2}{x^3}\right]$$

$$z = x^3\left[5 + 2\left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2\right] = x^3 f\left(\frac{y}{x}\right)$$

This is a homogeneous function of degree '3' in x and y.

(2) $u = x^5 \cos\left(\frac{y}{x}\right)$ is a homogeneous function of degree '5' in x and y.

Scanned with CamScanner

Euler's theorem:

If u is a homogeneous function of degree n in x and y, then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

Proof: Consider a homogeneous function of degree 'n' i.e, $u = x^n f(y/x) \rightarrow (1)$

partially differentiate (1) w.r.t 'x'

$$\frac{\partial u}{\partial x} = x^n\left[f'\left(\frac{y}{x}\right)\right]y\left(\frac{-1}{x^2}\right) + \left[f\left(y/x\right)\right]nx^{n-1}$$

Multiply by x

$$x\frac{\partial u}{\partial x} = x\left[-x^{n-2}yf'\left(\frac{y}{x}\right) + nx^{n-1}f\left(\frac{y}{x}\right)\right]$$

$$x\frac{\partial u}{\partial x} = -x^{n-2+1}yf'\left(\frac{y}{x}\right) + nx^{n-1+1}f\left(\frac{y}{x}\right)$$

$$x\frac{\partial u}{\partial x} = -x^{n-1}yf'\left(\frac{y}{x}\right) + nx^n f\left(\frac{y}{x}\right)$$

$$x\frac{\partial u}{\partial x} = -x^{n-1}yf'\left(\frac{y}{x}\right) + nu \rightarrow (2)$$

Scanned with CamScanner

partially differentiate ① w.r.t 'y'

$$\frac{\partial u}{\partial y} = x^n \left[f'\left(\frac{y}{x}\right)\right]\left(\frac{1}{x}\right) \quad (1)$$

Multiply by 'y'

$$y\frac{\partial u}{\partial y} = y x^{n-1} f'\left(\frac{y}{x}\right) \rightarrow (3)$$

② + ③ gives

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -x^{n-1} y f'\left(\frac{y}{x}\right) + nu + y x^{n-1} f'\left(\frac{y}{x}\right)$$

$$\Rightarrow \quad x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu \rightarrow (4)$$

Corollary: $x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} = n(n-1)u$

Proof: partially differentiate ④ w.r.t 'x'

$$x\left(\frac{\partial^2 u}{\partial x^2}\right) + \frac{\partial u}{\partial x}(1) + y\left(\frac{\partial^2 u}{\partial x \partial y}\right) = n\frac{\partial u}{\partial x}$$

Multiply by 'x'

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} + x\frac{\partial u}{\partial x} = nx\frac{\partial u}{\partial x}$$

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} = nx\frac{\partial u}{\partial x} - \frac{\partial u}{\partial x}$$

Scanned with CamScanner

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} = x\frac{\partial u}{\partial x}[n-1] \rightarrow (5)$$

Similarly partially differentiating (4) w.r.t 'y' and multiplying by 'y' we obtain $y^2\frac{\partial^2 u}{\partial y^2} + xy\frac{\partial^2 u}{\partial x \partial y} = y\frac{\partial u}{\partial y}[n-1] \rightarrow (6)$

(5) + (6) gives

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} + xy\frac{\partial^2 u}{\partial x \partial y}$$

$$= x\frac{\partial u}{\partial x}[n-1] + y\frac{\partial u}{\partial y}[n-1]$$

$$\Rightarrow x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} = [n-1]\left[x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right]$$

$$x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} = (n-1)nu$$

Scanned with CamScanner

Q1) If $u = \log\left(\frac{x^4+y^4}{x+y}\right)$, then show that

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$$

→ Given $u = \log\left(\frac{x^4+y^4}{x+y}\right)$

$$\exp u = \exp \log\left(\frac{x^4+y^4}{x+y}\right)$$

$$e^u = \frac{x^4+y^4}{x+y} = \frac{x^4\left(1+y^4/x^4\right)}{x\left(1+y/x\right)}$$

$$e^u = \frac{x^4}{x}\,\frac{\left[1+(y/x)^4\right]}{\left[1+(y/x)\right]}$$

$e^u = x^3 f(y/x)$ is a homogeneous function of degree 3 in x and y.

By Euler's theorem

$$x\frac{\partial}{\partial x}(e^u) + y\frac{\partial}{\partial y}(e^u) = 3e^u$$

$$xe^u\frac{\partial u}{\partial x} + ye^u\frac{\partial u}{\partial y} = 3e^u$$

$$e^u\left[x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right] = 3e^u \Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3.$$

Scanned with CamScanner

(Q2) If $u = \sin^{-1}\left(\frac{x^2y^2}{x+y}\right)$, then show that

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3\tan u$$

→ Given $u = \sin^{-1}\left(\frac{x^2y^2}{x+y}\right)$

$$\sin u = \frac{x^2y^2}{x+y}$$

$$\sin u = \frac{x^4\left(\frac{y^2}{x^2}\right)}{x\left(1+\frac{y}{x}\right)}$$

$$\sin u = \frac{x^4}{x}\frac{\left[\frac{y}{x}\right]^2}{\left(1+\frac{y}{x}\right)} = x^3 f\left(\frac{y}{x}\right)$$

$\sin u$ is a homogeneous function of degree 3 in x and y, by Euler's theorem

$$x\frac{\partial}{\partial x}(\sin u) + y\frac{\partial}{\partial y}(\sin u) = 3\sin u$$

$$x(\cos u)\left(\frac{\partial u}{\partial x}\right) + y(\cos u)\left(\frac{\partial u}{\partial y}\right) = 3\sin u$$

$$\cos u\left[x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right] = 3\sin u$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{3\sin u}{\cos u}$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3\tan u.$$

Scanned with CamScanner

(Q3) If $z = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, then prove that $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \sin 2u$.

→ Given $z = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$

$$\tan z = \frac{x^3+y^3}{x-y}$$

$$\tan z = \frac{x^3\left[1+\frac{y^3}{x^3}\right]}{x\left[1-\frac{y}{x}\right]} = \frac{x^2\left[1+\left(\frac{y}{x}\right)^3\right]}{\left[1-\left(\frac{y}{x}\right)\right]}$$

$\tan z = x^2 f(y/x)$ is a homogeneous function of degree 2 in x and y, by Euler's theorem,

$$x\frac{\partial}{\partial x}(\tan z) + y\frac{\partial}{\partial y}(\tan z) = 2\tan z$$

$$x\sec^2 z\frac{\partial z}{\partial x} + y\sec^2 z\frac{\partial z}{\partial y} = 2\tan z$$

$$\sec^2 z\left[x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}\right] = 2\tan z$$

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{2\tan z}{\sec^2 z}$$

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{2\sin z}{\cos z}\cos^2 z$$

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2\sin z\cos z$$

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \sin 2z$$

Scanned with CamScanner

classmate
Date
Page

Total Derivative:

① Let $z = f(x,y)$, $x = f_1(t)$, $y = f_2(t)$, the total derivative of 'z' w.r.t 't' is

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

② Let $z = f(x,y)$, $x = g_1(r,s)$, $y = g_2(r,s)$,

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r}$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial s}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

If $u = f(x,y,z)$ and $x = g_1(r,s)$ $y = g_2(r,s)$ and $z = g_3(r,s)$, then

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial u}{\partial z}\frac{\partial z}{\partial r}$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial u}{\partial z}\frac{\partial z}{\partial s}$$

Scanned with CamScanner

Differentiation of Implicit Functions :

If $f(x,y) = c$ be an implicit relation between x and y which defines as a differentiable function of x.

If $u = f(x,y)$, $x = f_1(t)$, $y = f_2(t)$

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} \rightarrow (1)$$

If $x = t$, then $\frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial y}\frac{dy}{dt} \rightarrow (2)$

Consider $f(x,y) = c$

$$0 = \frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx} \quad \text{from } (2)$$

$$\Rightarrow \frac{dy}{dx}\frac{\partial f}{\partial y} = -\frac{\partial f}{\partial x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-f_x}{f_y}$$ This formula is for the first differential coefficient of an implicit function.

Scanned with CamScanner

(Q) If $xy^3 - yx^3 = 6$ is the equation of a curve, then find the slope and equation of the tangent line at $(1,2)$.

→ Differentiate $xy^3 - yx^3 = 6$ implicitly w.r.t 'x'

$$x\,3y^2\frac{dy}{dx} + y^3(1) - \left[y\,3x^2 + x^3\frac{dy}{dx}\right] = 0$$

At $(1,2)$,

$$(1)3(2^2)\frac{dy}{dx} + 2^3 - \left[2(3)(1) + (1)\frac{dy}{dx}\right] = 0$$

$$12\frac{dy}{dx} + 8 - 6 - \frac{dy}{dx} = 0$$

$$(12-1)\frac{dy}{dx} + 2 = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{2}{11} = \text{slope}$$

Scanned with CamScanner

classmate
Date
Page

Equation of tangent line at (1,2) is

$$\frac{y-2}{x-1} = \frac{-2}{11}$$

$$(y-2)(11) = (-2)(x-1)$$

$$11y - 22 = -2x + 2$$

$$2x + 11y - 24 = 0$$

Scanned with CamScanner

(Q) If $u = 2e^x \sin y$, where $x = t^2$, $y = t-1$, compute $\frac{du}{dt}$ at $t=1$

→ $$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$$

$$\frac{du}{dt} = \frac{\partial}{\partial x}(2e^x \sin y)\frac{d}{dt}(t^2) + \frac{\partial}{\partial y}(2e^x \sin y)\frac{d(t-1)}{dt}$$

$$\frac{du}{dt} = (2\sin y)(e^x)(2t) + (2e^x)(\cos y)(1)$$

when $t=1$, $x=1$, $y=0$

$$\frac{du}{dt} = [(2\sin 0)(e^1)(2)] + [2e^1 \cos(0)]$$

$$\frac{du}{dt} = 2e$$

Q If $u = e^x \sin(yz)$, where $x = t^2$, $y = t-1$, $z = \frac{1}{t}$, compute $\frac{du}{dt}$ at $t=1$.

→ $$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}$$

$$\frac{du}{dt} = (\sin yz)e^x(2t) + e^x(\cos yz)(z)(1) + (e^x)(\cos yz)(y)\left(\frac{-1}{t^2}\right)$$

Scanned with CamScanner

When $t=1,\ x=1,\ y=0,\ z=1$

$$\frac{du}{dt} = \left[(\sin 0)(e)(2)\right] + \left[e(\cos 0)(1)\right] + \left[e(\cos 0)(0)(-1)\right]$$

$$\frac{du}{dt} = e$$

Q) If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, then show that $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$.

→ $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$

Let $u = f(r, s)$, $r = \frac{y-x}{xy} = \frac{y}{xy} - \frac{x}{xy}$

$$r = \frac{1}{x} - \frac{1}{y}$$

$$s = \frac{z-x}{xz} = \frac{z}{xz} - \frac{x}{xz} \Rightarrow s = \frac{1}{x} - \frac{1}{z}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\left(\frac{-1}{x^2}\right) + \frac{\partial u}{\partial s}\left(\frac{-1}{x^2}\right)$$

$$\frac{\partial u}{\partial x} = \frac{1}{x^2}\left[-\frac{\partial u}{\partial r} - \frac{\partial u}{\partial s}\right]$$

Scanned with CamScanner

$$x^2 \frac{\partial u}{\partial x} = -\frac{\partial u}{\partial r} - \frac{\partial u}{\partial s} \rightarrow (1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\left[0 - \left(\frac{-1}{y^2}\right)\right] + \frac{\partial u}{\partial s}(0)$$

$$\frac{\partial u}{\partial y} = \frac{1}{y^2}\frac{\partial u}{\partial r}$$

$$y^2 \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \rightarrow (2)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial z}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}\left[0 - \left(-\frac{1}{z^2}\right)\right]$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial s}\left(\frac{1}{z^2}\right)$$

$$z^2 \frac{\partial u}{\partial z} = \frac{\partial u}{\partial s} \rightarrow (3)$$

(1) + (2) + (3) gives

$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = -\frac{\partial u}{\partial r} - \frac{\partial u}{\partial s} + \frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}$$

$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$$

Scanned with CamScanner

(a) If $u = x^2 - y^3$, $x = 2r - 3s + 4$, $y = -r + 8s + 5$, find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial s}$.

→ $$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

$$\frac{\partial u}{\partial r} = (2x)(2) + (-2y)(-1) = 4x + 2y$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial s}$$

$$\frac{\partial u}{\partial s} = (2x)(-3) + (-2y)(8) = -6x - 16y$$

Scanned with CamScanner

(8) If $u = f(xz, \frac{y}{z})$, then show that

$$x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} - z\frac{\partial u}{\partial z} = 0.$$

→ Given $u = f(xz, \frac{y}{z})$.

Let $u = f(r, s)$

$r = xz, \quad s = \frac{y}{z}$

$$\frac{\partial r}{\partial x} = z, \quad \frac{\partial r}{\partial y} = 0, \quad \frac{\partial r}{\partial z} = x$$

$$\frac{\partial s}{\partial x} = 0, \quad \frac{\partial s}{\partial y} = \frac{1}{z}, \quad \frac{\partial s}{\partial z} = -\frac{y}{z^2}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}(z) + \frac{\partial u}{\partial s}(0)$$

$$\frac{\partial u}{\partial x} = z\frac{\partial u}{\partial r} \rightarrow (1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}\left(\frac{1}{z}\right)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial s}\left(\frac{1}{z}\right)$$

Scanned with CamScanner

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial z}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\, x + \frac{\partial u}{\partial s}\left(\frac{-y}{z^2}\right) \rightarrow (3)$$

$$x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} - z\frac{\partial u}{\partial z}$$

$$= x\left(z\frac{\partial u}{\partial r}\right) - y\left(\frac{1}{z}\frac{\partial u}{\partial s}\right) - z\left[x\frac{\partial u}{\partial r} - \frac{y}{z^2}\frac{\partial u}{\partial s}\right]$$

$$= xz\frac{\partial u}{\partial r} - \frac{y}{z}\frac{\partial u}{\partial s} - zx\frac{\partial u}{\partial r} + \frac{y}{z}\frac{\partial u}{\partial s}$$

$$= 0$$

Scanned with CamScanner